ECOM155: Asset Pricing, Trading and Portfolio Construction
Portfolio Assignment (80%)
January 2026

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Opens: Thursday, 11 December 2025, 9:00 AM
Due: Monday, 5 January 2026, 11:59 PM

Instructions:
This is an open-book portfolio project. You may refer to course materials or any other sources of information.

YOU CANNOT SUBMIT HANDWRITTEN ANSWERS. ANSWERS ARE TO BE SUBMITTED ON QMPLUS ONLY.

Answer ALL QUESTIONS. This portfolio assessment is set for 100 marks. However, the assessment counts as 80% towards the total course grade.

Upload a single file on QMPlus containing your answers to all questions, with the code appended at the end. Please UPLOAD ONLY ONE MICROSOFT WORD DOCX file. Please go to the ECOM155 module page, to the "Assessment" tab and you may submit under the area titled "Assessment 1 : Portfolio - 80%".

Overview

This portfolio assignment focuses on empirical asset pricing using the Fama-MacBeth two-pass regression framework. You will work with real industry portfolio data from Kenneth French's Data Library to test whether factor models can explain the cross-section of expected returns. This exercise combines theoretical understanding of asset pricing with practical implementation skills in R.

Data Files Provided:
You will use the following three CSV files provided on QMPlus:

1. **49_Industry_Portfolios.csv**: Monthly value-weighted returns for 49 industries from January 2000 to November 2024. The first column contains the date (YYYYMM format), and the remaining 49 columns contain monthly returns (in percentage) for each industry.

2. **Fama_French_Factors.csv**: Monthly Fama-French 3-factor returns and the risk-free rate. Columns are: date, Mkt_RF (market excess return), SMB (small minus big), HML (high

minus low), and RF (risk-free rate). All values are in percentage.

3. **Momentum_Factor.csv**: Monthly momentum factor returns. Columns are: date and Mom (momentum factor return in percentage).

Sample Period: All data files cover January 2000 to November 2024 (299 months).

Data Source: These files are cleaned versions of data from Kenneth French's Data Library (`https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`).

1 Data Preparation and Descriptive Statistics (20 Marks)

(a) Load the 49_Industry_Portfolios.csv file. Report the names of all 49 industries in your dataset. Verify that you have 299 months of data (January 2000 to November 2024). (5 Marks)

(b) Compute the average monthly return and standard deviation for each of the 49 industries over your sample period. Create a table showing the top 5 and bottom 5 industries ranked by average return. (5 Marks)

(c) Load the Fama_French_Factors.csv and Momentum_Factor.csv files. Merge these with your industry returns data. Report the mean, standard deviation, minimum, and maximum for each of the four factors (Mkt_RF, SMB, HML, Mom) over your sample period in a summary table. (5 Marks)

(d) Create a scatter plot with average industry returns on the y-axis and industry return volatility (standard deviation) on the x-axis. Does there appear to be a risk-return tradeoff? Discuss briefly. (5 Marks)

2 Time-Series Regressions (25 Marks)

The Fama-MacBeth procedure begins with time-series regressions to estimate factor loadings (betas) for each asset. For each of the 49 industry portfolios, run the following time-series regression:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i^{Mkt}(r_{M,t} - r_{f,t}) + \beta_i^{SMB} SMB_t + \beta_i^{HML} HML_t + \beta_i^{Mom} Mom_t + \varepsilon_{i,t}$$

where $r_{i,t}$ is the return on industry i in month t, $r_{f,t}$ is the risk-free rate, and $(r_{M,t} - r_{f,t})$ is the market excess return (Mkt_RF).

(a) Run the above regression for all 49 industries. Create a table showing the estimated factor loadings (β^{Mkt}, β^{SMB}, β^{HML}, β^{Mom}) and their t-statistics for 5 industries of your choice. (10 Marks)

(b) Report the R^2 values for all 49 industries. What is the average R^2 across all industries? What does this tell you about how well the four-factor model explains industry returns? (5 Marks)

(c) Create a histogram of the estimated market betas (β^{Mkt}) across all 49 industries. What is the range of market betas? Are there any industries with particularly high or low market exposure? (5 Marks)

(d) Identify the 5 industries with the highest momentum factor loadings (β^{Mom}) and the 5 industries with the lowest momentum factor loadings. Report these in a table with their estimated momentum betas and t-statistics. (5 Marks)

3 Cross-Sectional Regressions (30 Marks)

In the second pass of the Fama-MacBeth procedure, you run cross-sectional regressions each month to estimate factor risk premiums. For each month t, run the following cross-sectional regression across the 49 industries:

$$r_{i,t} - r_{f,t} = \lambda_0 + \lambda_1 \hat{\beta}_i^{Mkt} + \lambda_2 \hat{\beta}_i^{SMB} + \lambda_3 \hat{\beta}_i^{HML} + \lambda_4 \hat{\beta}_i^{Mom} + \eta_{i,t}$$

where $\hat{\beta}_i$ are the estimated betas from Question 2, and λ coefficients represent the factor risk premiums in month t.

(a) Run the cross-sectional regression for each month in your sample period (January 2000 to November 2024). Compute the time-series average of each estimated risk premium ($\bar{\lambda}_0$, $\bar{\lambda}_1$, $\bar{\lambda}_2$, $\bar{\lambda}_3$, $\bar{\lambda}_4$). (10 Marks)

(b) Test whether each average risk premium is statistically significant using the Fama-MacBeth t-statistics:

$$t(\bar{\lambda}_j) = \frac{\bar{\lambda}_j}{SE(\lambda_j)/\sqrt{T}}$$

where $SE(\lambda_j)$ is the standard deviation of λ_j across months and T is the number of months. Report your results in a table with the estimated risk premiums and their t-statistics. (10 Marks)

(c) Compare your estimated factor risk premiums with the actual average factor returns over the same period. Are they similar? What does this comparison tell you about whether the factors are priced in the cross-section? (5 Marks)

(d) Plot the time-series of the estimated market risk premium (λ_1) over your sample period. Are there any notable patterns or periods where the market risk premium was particularly high or low? Discuss. (5 Marks)

4 Long-Short Portfolio Strategy (25 Marks)

Using your estimated factor loadings from Question 2, construct a trading strategy based on industry characteristics.

(a) Based on the momentum factor loadings ($\hat{\beta}_i^{Mom}$) from Question 2, identify the 10 industries with the highest momentum betas (high momentum industries) and the 10 industries with the lowest momentum betas (low momentum industries). (5 Marks)

(b) Construct a long-short portfolio strategy: Go long (equal-weighted) the 10 high momentum beta industries and short (equal-weighted) the 10 low momentum beta industries. Rebalance this portfolio monthly using the most recent beta estimates. Compute the monthly returns of this long-short portfolio from January 2000 to November 2024. (10 Marks)

(c) Calculate the following performance metrics for your long-short portfolio:

- Average monthly return (annualized)
- Standard deviation (annualized)
- Sharpe ratio (assuming a risk-free rate of 0.0025 per month, or 3% annually)

- Maximum drawdown

Report these metrics and discuss whether this strategy appears profitable. (5 Marks)

(d) Run a factor regression of your long-short portfolio returns on the four factors (Mkt_RF, SMB, HML, Mom):

$$r_t^{LS} = \alpha + \beta^{Mkt}(r_{M,t} - r_{f,t}) + \beta^{SMB} SMB_t + \beta^{HML} HML_t + \beta^{Mom} Mom_t + \varepsilon_t$$

Report the alpha and its t-statistic. Is the alpha statistically significant? What does this tell you about the strategy's performance after adjusting for factor exposures? (5 Marks)

End of Examination. Good luck!